

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Artificial Intelligence and Data Science

ASSESSMENT TOOL 2 – SCENARIO BASED

Course Code:	DSA03	Course Title:	Natural Language Processing
CO Assessed:	CO2 – Manipulation	Assessment:	Assessment Tool 2 – SCENARIO BASED
Weightage:	35% of CIA		
CO5 Statement:	Design inflectional, derivational, finite-state parsing, stemming algorithms and for analyse morphological methods. (BL-3)		
Student Name:	Nirmal Kasi	Reg. No.:	192424248
Section / Batch:	B.Tech AI/DS	Date:	

Code of Conduct

I Nirmal Kasi (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: Nirmal

Instructions

- Read all questions carefully before attempting the answers.
- Answer all five questions. Each question carries equal marks.
- Show all intermediate steps, calculations, probability computations, tagging decisions, and justifications wherever applicable.
- Duration: 30 minutes.

Section / Topic	Focus	Q Nos
English Word Classes, Tagsets for English, Stochastic Part of Speech Tagging, HMM	Morphological parsing of disagree, agreement, and agreeable; identification of prefixes, suffixes, base forms, and semantic effects of derivation.	Q1

Annexure A – SCENARIO BASED

A company is developing an NLP-based grammar checker for educational applications. The system is trained using a manually POS-tagged corpus. Your task is to recreate the Hidden Markov Model (HMM) **without using any built-in NLP or HMM libraries**.

From the given corpus,

- determine the **Emission Probability**
- determine the **Transition Probability**
- implement the **Viterbi Algorithm**
- predict the POS tags for the new input sentence.

Use only Python dictionaries, loops, and basic programming constructs.

Training Corpus

Sentence	POS Tags
The/DT boy/NN eats/VBZ rice/NN	DT NN VBZ NN
The/DT girl/NN drinks/VBZ milk/NN	DT NN VBZ NN
A/DT cat/NN drinks/VBZ milk/NN	DT NN VBZ NN
The/DT dog/NN chases/VBZ cat/NN	DT NN VBZ NN
A/DT teacher/NN teaches/VBZ students/NNS	DT NN VBZ NNS
Students/NNS study/VBP English/NN	NNS VBP NN
Birds/NNS fly/VBP high/RB	NNS VBP RB
Children/NNS play/VBP games/NNS	NNS VBP NNS

New Input Sentence

The cat drinks milk

Tasks

1. Separate each sentence into words and POS tags.
2. Compute the **Emission Probability** of every word.
3. Compute the **Transition Probability** between POS tags.
4. Recreate the Hidden Markov Model using Python dictionaries.
5. Implement the **Viterbi Algorithm** without using any built-in HMM/NLP libraries.
6. Predict the POS tag sequence for “The cat drinks milk”

Q1. Separate each sentence into words and POS tags

1. The boy eats rice → DT NN VBZ NN
2. The girl drinks milk → DT NN VBZ NN
3. A cat drinks milk → DT NN VBZ NN
4. The dog chases cat → DT NN VBZ NN
5. A teacher teaches students → DT NN VBZ NNS
6. Students study English → NNS VBP NN
7. Birds fly high → NNS VBP RB
8. Children play games → NNS VBP NNS

POS Tags:

DT = Determiner

NN = Noun

VBZ = Verb, third-person singular present

NNS = Plural noun

VBP = Verb, non-third-person singular present

RB = Adverb

Q2. Emission Probability

Formula:

$$P(\text{word}|\text{tag}) = \text{Count}(\text{word}, \text{tag}) / \text{Count}(\text{tag})$$

Tag counts:

$$\text{DT} = 7$$

$$\text{NN} = 9$$

$$\text{VBZ} = 5$$

$$\text{NNS} = 4$$

$$\text{VBP} = 3$$

$$\text{RB} = 1$$

Emission probabilities:

$$\text{The} | \text{DT} = 5/7 = 0.7143$$

$$\text{A} | \text{DT} = 2/7 = 0.2857$$

$$\text{boy} | \text{NN} = 1/9 = 0.1111$$

$$\text{girl} | \text{NN} = 1/9 = 0.1111$$

$$\text{rice} | \text{NN} = 1/9 = 0.1111$$

$$\text{cat} | \text{NN} = 2/9 = 0.2222$$

$$\text{milk} | \text{NN} = 2/9 = 0.2222$$

$$\text{dog} | \text{NN} = 1/9 = 0.1111$$

$$\text{teacher} | \text{NN} = 1/9 = 0.1111$$

$$\text{English} | \text{NN} = 1/9 = 0.1111$$

eats | VBZ = $1/5 = 0.2$
drinks | VBZ = $2/5 = 0.4$
chases | VBZ = $1/5 = 0.2$
teaches | VBZ = $1/5 = 0.2$

students | NNS = $1/4 = 0.25$
Students | NNS = $1/4 = 0.25$
games | NNS = $1/4 = 0.25$
Children | NNS = $1/4 = 0.25$

study | VBP = $1/3 = 0.3333$
fly | VBP = $1/3 = 0.3333$
play | VBP = $1/3 = 0.3333$

high | RB = $1/1 = 1$

For the new sentence:

The $\rightarrow$ DT $\rightarrow 5/7$
cat $\rightarrow$ NN $\rightarrow 2/9$
drinks $\rightarrow$ VBZ $\rightarrow 2/5$
milk $\rightarrow$ NN $\rightarrow 2/9$

Q3. Transition Probability

Formula:

$$P(\text{tag2}|\text{tag1}) = \text{Count}(\text{tag1}, \text{tag2}) / \text{Count}(\text{tag1})$$

$$\text{DT} \rightarrow \text{NN} = 7/7 = 1.0$$

$$\text{NN} \rightarrow \text{VBZ} = 5/7 = 0.7143$$

$$\text{NN} \rightarrow \text{END} = 2/7 = 0.2857$$

$$\text{VBZ} \rightarrow \text{NN} = 4/5 = 0.8$$

$$\text{VBZ} \rightarrow \text{NNS} = 1/5 = 0.2$$

$$\text{NNS} \rightarrow \text{VBP} = 3/4 = 0.75$$

$$\text{NNS} \rightarrow \text{END} = 1/4 = 0.25$$

$$\text{VBP} \rightarrow \text{NN} = 1/3 = 0.3333$$

$$\text{VBP} \rightarrow \text{RB} = 1/3 = 0.3333$$

$$\text{VBP} \rightarrow \text{NNS} = 1/3 = 0.3333$$

$$\text{RB} \rightarrow \text{END} = 1$$

Q4. HMM using Python dictionaries

```

emission = {
  "DT": {"The": 5/7, "A": 2/7},
  "NN": {
    "boy": 1/9,
    "girl": 1/9,
    "rice": 1/9,
    "cat": 2/9,
    "milk": 2/9,
    "dog": 1/9,
    "teacher": 1/9,
    "English": 1/9
  },
  "VBZ": {
    "eats": 1/5,
    "drinks": 2/5,
    "chases": 1/5,
    "teaches": 1/5
  },
  "NNS": {
    "students": 1/4,
    "Students": 1/4,
    "games": 1/4,
    "Children": 1/4
  },
  "VBP": {
    "study": 1/3,
    "fly": 1/3,
    "play": 1/3
  },
  "RB": {"high": 1}
}

```

```

transition = {
  "DT": {"NN": 1.0},
  "NN": {"VBZ": 5/7, "END": 2/7},
  "VBZ": {"NN": 4/5, "NNS": 1/5},
  "NNS": {"VBP": 3/4, "END": 1/4},
  "VBP": {"NN": 1/3, "RB": 1/3, "NNS": 1/3},
  "RB": {"END": 1.0}
}

```

Q5. Viterbi Algorithm

The Viterbi algorithm finds the most probable sequence of POS tags.

```
words = ["The", "cat", "drinks", "milk"]
```

```
tags = ["DT", "NN", "VBZ", "NNS", "VBP", "RB"]
```

```
viterbi = []
```

```
backpointer = []
```

```
first = {}
```

```
first_back = {}
```

```
for tag in tags:
```

```
    if words[0] in emission.get(tag, {}):
```

```
        first[tag] = emission[tag][words[0]]
```

```
        first_back[tag] = None
```

```
viterbi.append(first)
```

```
backpointer.append(first_back)
```

```
for i in range(1, len(words)):
```

```
    current = {}
```

```
    current_back = {}
```

```
    for tag in tags:
```

```
        if words[i] not in emission.get(tag, {}):
```

```
            continue
```

```
        best_probability = 0
```

```
        best_previous = None
```

```
        for previous_tag in viterbi[i-1]:
```

```
            if tag in transition.get(previous_tag, {}):
```

```
                probability = (
```

```
                    viterbi[i-1][previous_tag]
```

```
                    * transition[previous_tag][tag]
```

```
                    * emission[tag][words[i]]
```

```
                )
```

```
                if probability > best_probability:
```

```
                    best_probability = probability
```

```
                    best_previous = previous_tag
```

```
        if best_previous is not None:
```

```
            current[tag] = best_probability
```

```
            current_back[tag] = best_previous
```

```
viterbi.append(current)
```

```
backpointer.append(current_back)
```

```
best_tag = max(viterbi[-1], key=viterbi[-1].get)
```

```
result = [best_tag]
```

```
for i in range(len(words)-1, 0, -1):
```

```
    best_tag = backpointer[i][best_tag]
```

```
    result.append(best_tag)
```

```
result.reverse()
```

```
print(result)
```

Q6. Prediction for "The cat drinks milk"

The → DT

cat → NN

drinks → VBZ

milk → NN

Final POS tag sequence:

DT NN VBZ NN

Therefore:

The/DT cat/NN drinks/VBZ milk/NN

Rubrics for Calculation

Criteria	Weightage (%)	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Understanding of Scenario	20%	Demonstrates clear and complete understanding of the scenario and context	Shows good understanding with minor gaps	Partial understanding; misses key aspects	Misinterprets or fails to understand the scenario
Identification of Key Issues	20%	Accurately identifies all relevant issues and factors involved	Identifies most key issues with minor omissions	Identifies limited issues; some irrelevant points	Unable to identify key issues
Application of Concepts	25%	Applies appropriate concepts effectively to address the scenario	Applies concepts with minor errors	Limited or partially correct application	Unable to apply relevant concepts
Decision Making	20%	Makes well-reasoned, logical decisions with strong rationale	Makes reasonable decisions with some justification	Makes weak decisions with limited justification	No clear decisions made or rationale provided
Clarity of Response	15%	Response is clear, structured, and logically presented	Generally clear with minor issues	Somewhat unclear or poorly structured	Disorganized and difficult to understand

Q. No. / Criteria Level	Understanding of Scenario	Identification of Key Issues	Application of Concepts	Decision Making	Clarity of Response	Sub. Total	Total %
1	4	4	4	4	4	20	90
2	4	4	4	4	4	20	
3	3	3	3	3	3	15	
4	4	4	4	4	4	20	
5	4	4	4	4	4	20	

Faculty In-charge	Course Coordinator	Program Director
Signature: _____	Signature: _____	Signature: _____
Date: _____	Date: _____	Date: _____